CSK DEVELOP3RS

23441

PRESENTS



KHRONOS

DECODE 2025-2026 | SOCAL CHAMPIONSHIP

POWERED BY:

REV ROBOTICS

HAAS Gene Haas Foundation

SOFTBOX LLC

goBILDA DREAM • MAKE • REPEAT

onshape®

FIRST®

4aiSoft

CSK FOUNDATION

INTUITIVE Foundation

CNC madness

Our Story

We are CSK Develop3rs 23441, a **student-led FTC Robotics** team based in Cerritos, California. Our team is entirely run by students focusing on innovation, **teamwork**, and STEM outreach. We are dedicated to creating competitive robots, **fostering a love for engineering**, and programming along our journey. FIRST has made a deep and far impact in our team's community, so our team has a mission to spread STEM and FIRST to communities **across the globe!**

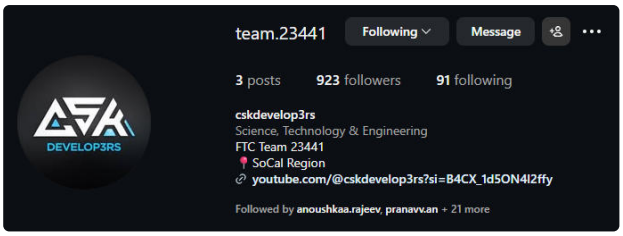


Our Team Values

- Always strive for technical excellency
- Promote **STEM** and ***FIRST*** in our **local community** and **across the world** (because everyone deserves access to STEM)
- Never forget to **have fun!**

Social Media Presence

- Showcased on multiple FTC youtube channels with large followings such as **FUN Robotics Network** and **Brogan Pratt**.
- Expanded over social media—received over **30,000 views** on our first youtube video of the season, and gained **900+ followers** on Instagram, while connecting with several teams over discord.



Risk Management

We apply principles from **FMEA** (Failure Mode and Effects Analysis) and **design matrices** to systematically evaluate risks, make informed design decisions, and prevent failures in matches.

Weights: 1 = not important 5 = super important

	x5	x3	x4	x2	x4	x90
Design	Vectoring	Speed	Precision	Weight	Reliability	Total
Idea 1	1	4	2	1	3	39
Idea 2	3	4	3	1	3	53
Idea 3	4	2	4	3	2	56
Idea 4	5	4	3	2	3	65
Idea 5	5	4	4	3	4	75

Intake Design Matrix

Failure Mode and Effects Analysis (FMEA)

We use FMEA to prevent match failures *before they happen*.

- Severity: How serious?
- Occurrence: How likely?
- Detection: Can we catch it?

Risk Priority Number (RPM): S × O × D

Judging Categories Highlights — INSPIRE

- In addition to **exceeding the standards for the each judged award** our team has become a **true ambassador of *FIRST*** and ***Gracious Professionalism®***
- Throughout our three years as a team, we've brought industry professionals and STEM access advocators together with *FIRST* teams as part of our **mission to bring quality STEM access to all people, regardless of their background**
- Although we are fierce competitors and always strive for excellency, we are pioneers of **"if you want to travel fast, go alone; if you want to travel far, go together."** We always offer a helping hand to other teams, and graciously accept help from others.

THINK (Throughout Portfolio)

- **Design Matrixes, FMEA, Constant Iteration/Prototyping**
- **Full documentation of CAD Refinement over the season**
- **Calculus-based calculations for optimization of pivot arm**

REACH/CONNECT (Page 2 - 5)

- **Mentors from Amazon Web Services, Walmart Global Tech, Elevance Health, Mercedes and other FIRST teams.**
- **Promoted STEM within our community with local engagement events and had an impact on 5000+ members.**

INNOVATE (Page 6-10)

- **Fully custom pocketed carbon fiber drivetrain**
- **Unique 360 degrees rotating turret**
- **One of a kind lift system for endgame**
- **Optimal Anti-Defense Ramp System**

DESIGN (Page 6-10)

- **Aesthetic Pocketing on Sideplates & Turret**
- **Industrial techniques through our tolerancing in our turret, CAD/iteration process, and unique mechanisms**
- **Active Intake featuring multicolor prints, rigid mounting and support, and intake consisting**

CONTROL (Pages 11 - 13)

- **Developed ARC-CV which is a team original CameraVision method.**
- **Utilized Vision Portal involving a fully custom vision pipeline, selection algorithm, and robot movement algorithm.**

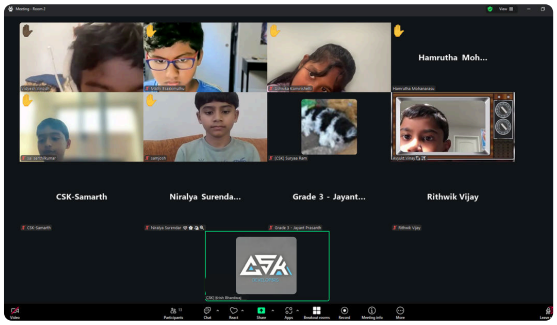
SUSTAIN (Pages 14-15)

- **Determined to establishing longevity through our team**
- **Managing our finances successfully**
- **Defining a clear order of structure within our team**
- **Mentoring younger shadow members for the future**

Global STEM Access Mission

Our team is committed to **expanding access to STEM education** for students who lack exposure to robotics and engineering opportunities. Through **virtual outreach across India,** we eliminate barriers such as cost, transportation, and limited local resources by hosting interactive Zoom-based robotics sessions.

By using **technology**, we ensure that opportunity is not limited by **geography, income, or access to local resources**, allowing students from underserved communities to explore robotics and STEM regardless of their circumstances.



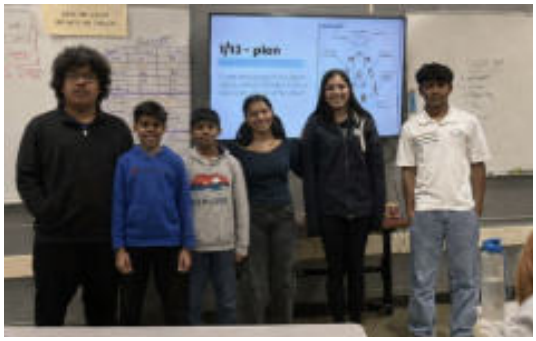
Empowering First-Generation STEM Learners



Our outreach also focuses on **mentorship and guidance** for students from **underserved and low-income backgrounds**. Many of the students we work with are learning about robotics for the first time, so our team emphasizes **encouragement and support**. By sharing our own experiences in STEM and robotics, we help students understand that these opportunities are **accessible and achievable**, even if they are just beginning their journey.

Exposure to Global STEM Pathways

Many students we work with are unaware of international robotics competitions or engineering career paths. We introduce them to FTC, innovation awards, research opportunities, and STEM scholarship pathways.
By showing real-world applications and future opportunities, we expand their academic vision beyond their immediate environment. Awareness creates ambition.



Team Impact

Through our international virtual outreach, we have:

- Reached students across **multiple** regions in India
- Eliminated geographic and financial barriers through Zoom-based **access**
- Introduced first-generation learners to FTC and robotics pathways
- Built **international** mentorship connections

100% Free Access

25+ Virtual Sessions Hosted

20+ Workshop Sessions

12+ STEM Topics Covered

200+ Students Reached

4+ REGIONS Across India

In addition to virtual mentorship, we host **introductory robotics workshops** that focus on **curiosity, creativity, and learning** rather than competition. These sessions help students build foundational knowledge of robotics and engineering while developing **confidence** in their abilities. Through our international outreach efforts, we aim to create **pathways for underprivileged students** to explore STEM and become more aware of opportunities available through programs like FTC.



CAD WORKSHOPS

Ten middle school students participated in our three-week CAD workshop, where they gained hands-on experience in computer-aided design using platforms such as Fusion 360, AutoCAD, and TinkerCAD. Throughout the program, students learned how to create 2D sketches, develop 3D models, and understand the fundamentals of engineering design. By the end of the workshop, participants were able to design their own original components, apply measurements and constraints accurately, and think critically about how digital designs translate into real-world structures.

Student-Led Micro Projects

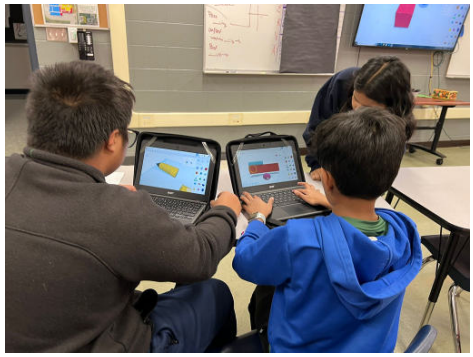
Beyond structured lessons, students complete small independent robotics challenges where they apply what they've learned. These micro projects include basic obstacle-avoidance simulations, simple sensor-based programs, and engineering design sketches.

Students present their solutions at the end of each cycle, strengthening public speaking skills and technical communication. This transforms them from passive learners into active problem-solvers.



Peer Collaboration

We connect students from different regions to work on shared engineering challenges. Through breakout discussions and collaborative design tasks, students learn teamwork, idea exchange, and structured brainstorming.
This builds leadership and collaboration skills while exposing them to diverse perspectives and problem-solving approaches.
STEM becomes not just technical training, but a platform for teamwork and communication.



Community Outreach Mission

Our team actively promotes STEM in our local community by hosting **robot demonstrations, interactive workshops,** and public STEM events. We focus on making robotics approachable and engaging for students of all ages, especially those who may not have prior exposure to engineering. Through **hands-on interaction** and **real-world applications,** we show that STEM is **accessible, creative, and impactful.**



1st Annual CSK STEM Festival

In addition, our team hosted the **first annual CSK STEM Festival** to further expand STEM access in our community. We organized this event with **a variety of different STEM-oriented organizations** to give children the opportunity to explore many fields of STEM in one place. The festival created an engaging and inclusive learning environment and reached over **200 community members**, helping promote long-term interest in STEM education.

Community Impact

Through our outreach efforts, we have:

- Hosted **large-scale** community STEM events
- Engaged **familes** and **first-time learners**
- Promoted **hands-on robotics experiences**
- Encouraged students to pursue engineering pathways

5,000+

COMMUNITY MEMBERS ENGAGED

Hands-On STEM Engagement

We prioritize **interactive learning over passive observation**. At events like the Cerritos Science & Art Expo, students engage directly with our robot, participate in demonstrations, and ask questions about how robotics works. By allowing students to learn through experience, we help build **confidence, curiosity, and long-term interest** in engineering. By demonstrating how robotics addresses meaningful challenges, students begin to see STEM as more than a subject—they see it as a **tool for change.**

Cerritos Library & Community Events

Hands-on interaction is a key part of our outreach efforts. Rather than having students only observe, we encourage them to actively explore STEM **through direct interaction** with our robot. At the Cerritos Science & Art Expo, students were able to participate in activities that allowed them to **learn by doing**, which helped strengthen their **curiosity and confidence** in STEM.

Community STEM Awareness

To expand participation beyond single events, our team launched a focused STEM awareness campaign across the community. We designed informational flyers, shared digital announcements, and partnered with local schools and libraries to promote upcoming robotics workshops and demonstrations. By increasing visibility, we reached families who may not typically seek out STEM programs, boosting attendance and introducing new students to accessible engineering opportunities.



Expanding Access to Enginnering

Many students in our community have limited exposure to robotics and engineering careers. To address this gap, our team brings accessible, hands-on STEM activities directly to local schools and public spaces. Students experiment with real robot components, test simple coding logic, and explore how sensors and motors work together. We intentionally design activities that are beginner-friendly while still challenging. Whether a student is completely new to STEM or already curious about robotics, they leave with a stronger understanding of how engineering solves real-world problems.



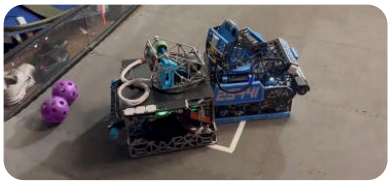
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STEM becomes not just technical training, but a platform for teamwork and communication.

On average, FTC teams compete in about **2 to 3 events** per season. We work to expand access to FIRST experiences beyond competition through community events, mentorship, and online outreach.

DEVELOP3RS Preseason Invitational

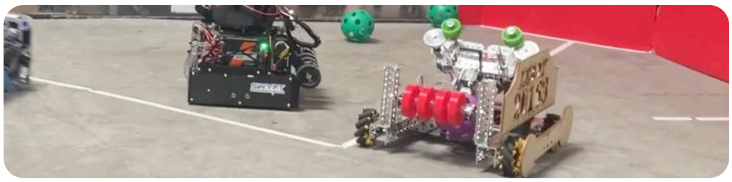
We have participated in and helped **organize** a **FTC scrimmage** for local teams, providing drivers with additional match practice before official league meets. Teams used this opportunity to test autonomous routines, refine mechanisms, and scrimmage in a **low-pressure** environment before competition.





Socal FTC "Open Gym"

Throughout the season and offseason, we have **hosted work sessions** where FTC teams could bring their robots to practice, troubleshoot issues, and **collaborate** with peers alongside our team members. These sessions helped teams stay **engaged** and **improve** performance outside of official events.



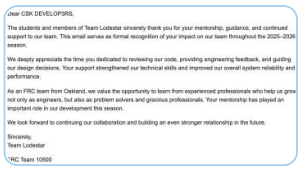
FIRST Mentorship

Over multiple seasons in FIRST programs, we have developed strong technical and leadership skills through designing, building, programming, and iterating competitive robots. As student leaders, we have learned how to manage projects, mentor younger members, communicate with judges and sponsors, and solve real engineering problems under time pressure. However, our growth is not just for our own success. We are committed to sharing our knowledge to expand access to STEM in our community and beyond by mentoring new teams, hosting workshops, supporting FLL and FTC programs, and guiding students through CAD, programming, and outreach planning. By empowering others with the skills and confidence to participate, we aim to create lasting growth in FIRST and inspire the next generation of innovators.

Our Purpose
We work to make robotics accessible, scalable, and sustainable for every student we reach.

Mentorship Model

We follow a simple cycle of growth: learn, apply, and multiply. As we develop advanced skills in programming, mechanical design, and vision systems, we immediately convert that knowledge into structured lessons and mentorship opportunities. This approach allows our impact to scale each season as mentored teams go on to support others in their own communities.


Dear CSK DEVELOP3RS,

The students and members of Team Lodestar sincerely thank you for your mentorship, guidance, and continued support to our team. This email serves as formal recognition of your impact on our team throughout the 2025–2026 season.

We deeply appreciate the time you dedicated to reviewing our code, providing engineering feedback, and guiding our design decisions. Your support strengthened our technical skills and improved our overall system reliability and performance.

As an FRC team from Oakland, we value the opportunity to learn from experienced professionals who help us grow not only as engineers, but also as problem solvers and gracious professionals. Your mentorship has played an important role in our development this season.

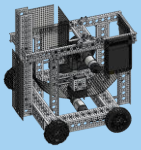
We look forward to continuing our collaboration and building an even stronger relationship in the future.

Sincerely,
Team Lodestar
FRC Team 10500

Above: **Certification of Mentorship** of **FRC** Team in our community

FIRST Teams Mentored
2 Local FLL Teams
4 FTC Teams from California, Virginia, Brazil, and India
1 International FRC Team
150+ More Assisted

FTC 13201 Team Hazmat
Assisted with CAD using Onshape, design choices, Tele-op automations, Using path following libraries for autonomous.



FTC 19510 VolTech
Worked with the team to build and explain an OpenCV vision pipeline for sample detection and contour tracing.

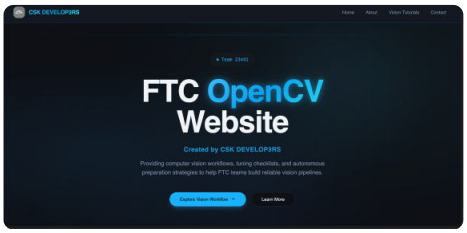


FTC 3565 Ghost Robotics
Assisted with CAD + design choices and offered software improvements focused on reliability, tuning, and match performance.



FRC 10500 Lodestar
Created a tutorial and assisted with Limelight calibration, detection tuning, and telemetry visualization.





We created an FTC OpenCV website to **support** the teams we **mentor** with computer vision development. Through our experience in FIRST Tech Challenge, we realized that many newer teams **struggle** with vision pipelines, camera calibration, contour detection, and integrating OpenCV into autonomous code. To make this easier, we **built** a structured online resource that explains key concepts in simple steps, includes example code snippets, troubleshooting tips, and visual diagrams, and walks teams through setting up and testing their pipelines. By **turning** our trial-and-error experience into **clear** documentation, we help other teams **save time**, avoid common mistakes, and **confidently** implement vision systems into their robots.

Spreading FIRST Online

We actively participate in online FIRST communities, answering technical questions, reviewing CAD files, and sharing programming resources to help teams improve throughout the season.

- Auton Sample Codes
- CAD File Review
- Onshape Library of Parts
- Teleop Sample Codes

Collection of Resources for FIRST Teams created by our team.

Outreach Stats

25+ FIRST Teams	**35+** Events Hosted	**45K+** Total People Reached	**500+** Outreach Hours
8+ Countries	**6+** States	**25+** STEM Organizations	**35K+** Social Media Views

Connecting with Experts

Our team is fully **student-run**, but as a garage based team, we rely on mentors to help guide us towards our goals. Our **network of mentors** include industrial **professionals** in the healthcare, finance, engineering, computer science, and business fields. We also focus on **building connections within the STEM community** through outreach events.

Industrial Mentors



Kiran C.
Senior Software Director at *Elevance Health*
Connected Through: FTC Community
Assisted With: Robot software architecture and system reliability



Lingareddy N.
Senior Engineering Leader at *Walmart Global Tech*
Connected Through: Industry Mentor
Assisted With: Engineering design reviews and risk mitigation



Goutham N.
Senior Leader at *Amazon Web Services*
Connected Through: FTC Community
Assisted With: Scalable telemetry, data logging, and system planning



Murugan A.
Senior Navigation Development Engineer at *Mercedes*
Connected Through: Industry Mentor
Assisted With: Autonomous programming logic and sensor integration

How Mentorship Improves Our Robot

Before Mentorship	Mentor Intervention	After Improvements
• Inconsistent auto	• Design reviews	• Stable autonomous
• Messy wiring	• Coding advice	• Optimized turret control
• Limited testing	• System Feedback	• Competition readiness

Acquiring Mentors

We find our mentors through many different mediums, such as through our FTC community, cold emailing small businesses in our neighborhoods, and connecting with industry professionals.

Training Tuesdays

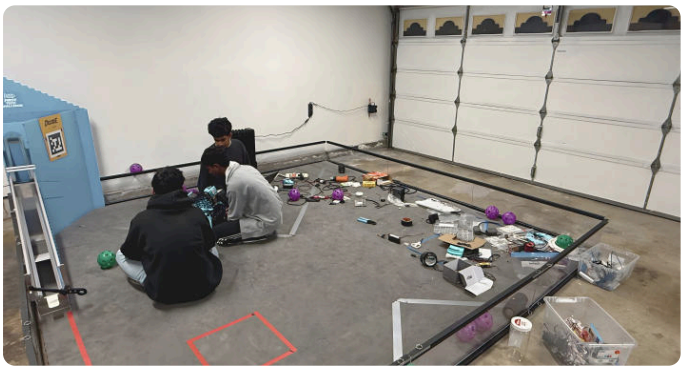
Dedicated **2**-hour training sessions held every Tuesday feature mentors from **various** industries who share **real-world** workforce skills such as project management, technical communication, structured problem solving, and professional collaboration, helping us better **understand** industry expectations and directly **apply** those principles to robotics design, teamwork, and competition performance.

September
- **Engineering Design Reviews**
- Conducted structured mentor led design critiques
- Reorganized wiring for cleaner power distribution
- Refactored autonomous code for stability

November
- **Mechanism Optimization**
- Developed rotating turret with mentor support
- Integrated sensors to enhance scoring consistency
- Assessed risk before full implementation

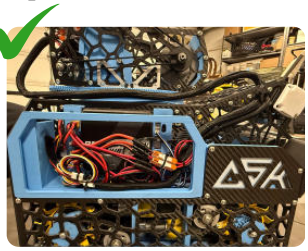
January
- **Competition Enhancement**
- Implemented mentor-recommended autonomous improvements
- Fine-tuned turret software for higher accuracy

We would like to thank our mentors for our success in the 2025-2026 Decode Season.



Key Example #1





Problem: Our robot experienced inconsistent, autonomous movement due to unstable software logic and inefficient wiring practices.
Solution: With **guidance** from our **mentors**, we cleaned up our wiring practices and restructured our autonomous code to **better optimize our runs.**

Key Example #2

Our team designed a **rotating turret** to allow flexible scoring angles without repositioning the drivetrain.



Lingareddy N: Advised on material selection and risk factors for each component type

Murugan A: Provided guidance on sensor feedback integration and signal filtering methods

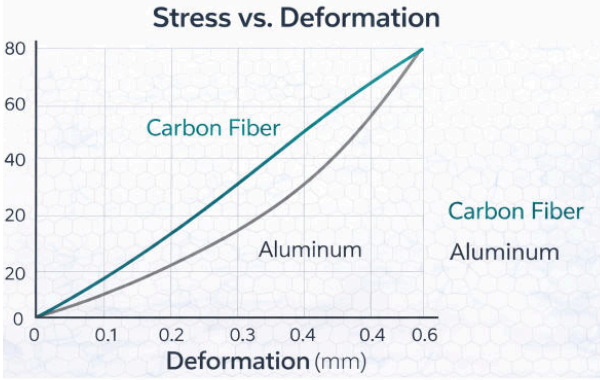
Goutham N: Reviewed wiring layout for an optimal and reliable autonomous run

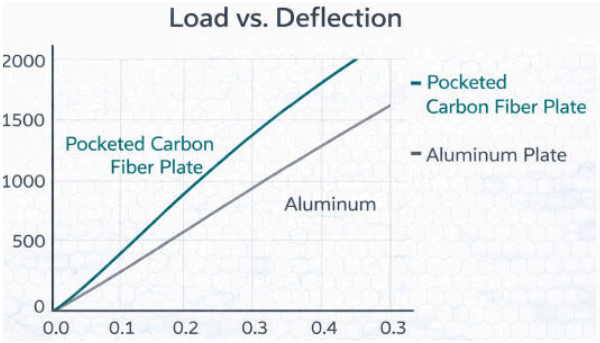
Our team has connected with 50+ STEM organizations through mentorship, collaboration, and outreach. Additional details are available in our pit display.

Carbon Fiber Structural Testing

To reduce weight while **maintaining** rigidity, we used pocketed carbon fiber plates instead of traditional aluminum sideplates. Carbon fiber **provides** a higher strength-to-weight ratio, allowing the drivetrain to **remain** rigid during impacts, defense, and rapid acceleration.

Through **testing** and **comparison** with aluminum, carbon fiber showed higher stiffness while significantly **reducing** weight, **improving** drivetrain stability, efficiency, and overall robot performance during **high-speed** movement.





Material Stress Graph Analysis

Using computational analysis **tools** and graph **modeling**, we created stress vs deformation curves to visualize how each material behaves under increasing load.

Key results from our graph analysis:
- Carbon fiber **maintained** lower deformation at higher stress levels
- Aluminum reached structural flex earlier under identical loading conditions
- Reduced deformation **improves** drivetrain alignment and gear efficiency

These results **reinforced** our design choice to use pocketed carbon fiber plates for our drivetrain superstructure.

Dual-Camera Vision System

Our robot uses a **dual-camera vision system** to improve field awareness and artifact detection.

- Camera 1 – AprilTag Localization: used for robot positioning and scoring alignment.
- Camera 2 – Artifact Detection: used to detect and track artifacts during autonomous.

Separating these tasks allows the robot to **localize** and **detect** objects at the same time, **improving** autonomous reliability and response speed.



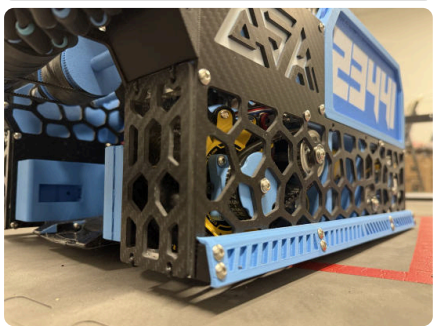
This camera is used for tracking the goal using AprilTags

This Limelight 3A is used for intaking artifacts clusters during autonomous

Endgame: Tilt Mechanism

Our tilt mechanism uses a slider mounted on an MGN linear rail, actuated by an over-centering linkage. This over-centering design allows the robot to remain securely tilted even after the program ends and the servos lose power at the conclusion of the match. To further improve robustness, we added omni wheels to the bottom of the tilt assembly. This addition allows the robot to maintain mobility and drive while tilted, while also preventing damage to the field tiles.

In Game Demo: Our robot is parked in the endgame zone and the ability to tilt provides space for the other team in our alliance to park together

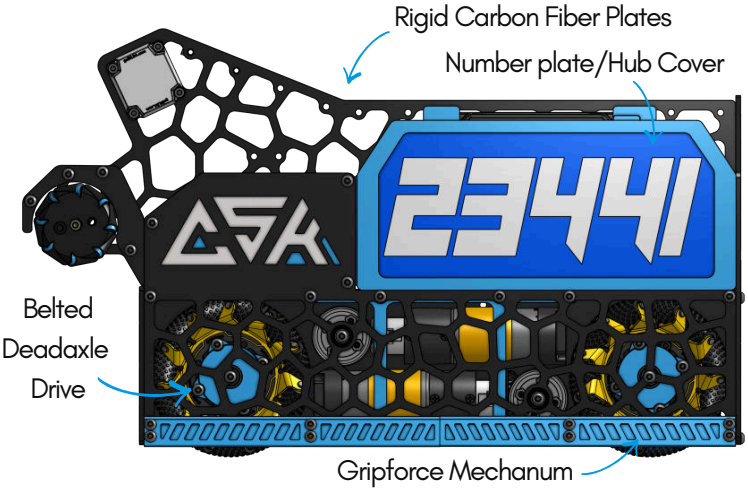


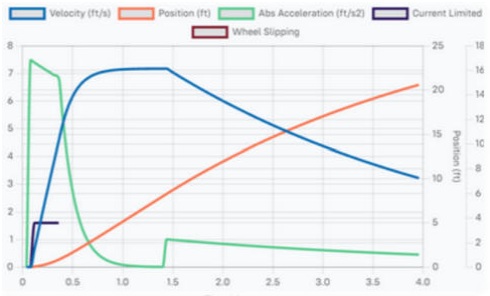


Our lifts are carefully constructed with slides, servos, and odometry pods which provides a strong structural basis for our robot to rest it's weight on while also successfully minimizing space

Drivetrain & Electrical

Drivetrain Overview





- **1.46s** 8ft Sprint Dist.
- **7.17ft/s** Max Speed
- **0.7** Coef. of Static Friction
- Few Failure Points
- Grip-Force Traction
- $500 Cost
- 70.7% Power in Strafe

Design Consideration: Swerve

- Full Speed (4-way)
- +60% Pushing Power
- More Failure Points
- $1200 Cost
- 2.5A+ Current Draw



Key Features

- 14.4:1 Ratio optimized for acceleration over short distances (1.46s for 8ft. Sprint Distance)
- Compact Parallel Plate Bevel Drive
- 1/8" Pocketed CF. Plate Superstructure for Rigidity (Pocketing Saves ~50% Weight)
- 14"x14" & Low COG for stable movement
- Fully Designed in Onshape CAD Software

V1: 17 Tooth Pulley



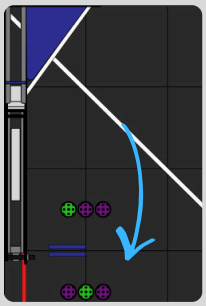
440 RPM
13:6:1 Ratio

V2: 18 Tooth Pulley



416 RPM
14:4:1 Ratio

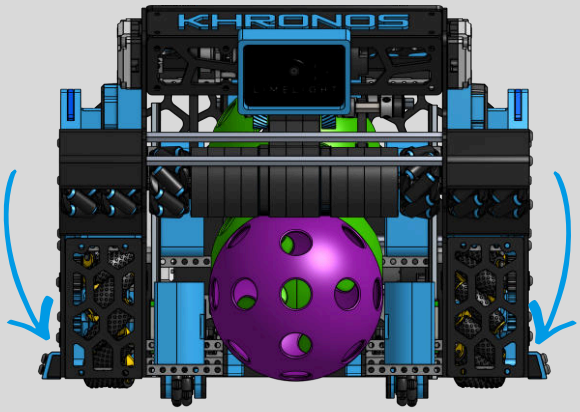
Our drivetrain's acceleration was too slow for the artifact cycle distances, so we used the ILITE Calculator to adjust the gear ratio from 13.6:1 to 14.4:1, ultimately saving 2s per cycle.



Problem: High arm acceleration and inertia cause slight tipping.

Solution: Added wedges on the sides to stabilize when tipping begins.

Result: Increased by 2 cycles per 120 seconds!



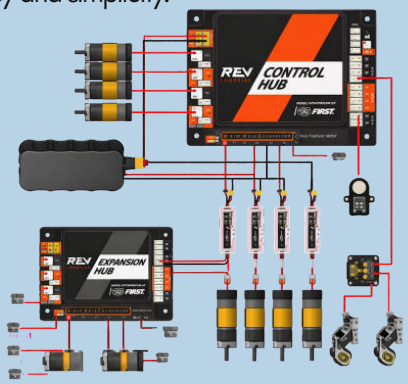
Serviceability

- Packed all of our wires into the cable sleeves to prevent any disconnections but also gave a cleaner look
- A separate spare for each motor/servo is made as spare for competition
- Each vulnerable component & 3d printed part is has a spare for competition

Electronics: Schematic Diagram

Our electronics system is built around a REV Control Hub paired with a REV Expansion Hub, eliminating the need for a separate REV Servo Hub. All motors, sensors, and encoders are distributed directly across the two hubs to maximize efficiency and simplicity.

- More compact and organized wiring layout
- Reduced overall system complexity
- Faster communication between hardware and control loops
- Improved reliability through fewer points of failure

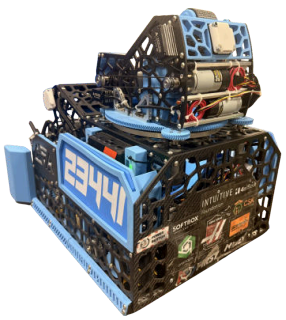


Industrial Design

- Aesthetic Consistency: A sleek, black and blue design for a professional appearance while matching team colors with LEDs
- Brand Integration – Logos and sponsor stickers enhance visibility and proudly showcase them

Logos clearly labeled throughout robot

Side skirts for style and protection

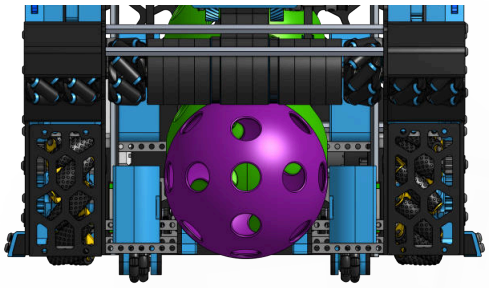


LEDs to make our robot stand out

Sponsorships clearly labeled

Intake and Transfer

Intake/Transfer Overview



For this season's intake, we prioritized a **touch-it-own-it** style design, with efficiency and speed as our **primary goals.** After more than **eight total iterations** and refinements, we finalized our **V3 intake**, which has proven to be highly **reliable and effective**. The intake spans the **full width** of the robot and uses **vectored wheels** on both sides to guide artifacts smoothly **toward the center.** Running at approximately 2,000 RPM, it allows for **near-instant** artifact intake. The first **three** intake stages are powered, while the transfer system is powered by **separate motor.** The transfer system consists of **three stages,** including a **back roller,** all operating at **2,500 RPM,** enabling consistent and **rapid firing.** Through **extensive iteration** and testing, this design has shown **strong** performance and reliability, and we have **not** encountered issues since **finalizing** it.

Intake Version 1 (M0 and M1)



- Good artifact compression
- Simple to assemble
- Slow RPM (435)
- Cannot intake agaisnt the gate
- Narrow intaking area

Intake Version 2 (M2 and M3)



- Good artifact compression
- Decent RPM (1150)
- Wider intaking area
- Cannot intake against the gate
- Jams while vectoring two balls

Intake Version 3 (Final)

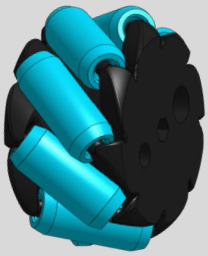


- Good artifact compression
- Fast RPM (2000)
- Wider intaking area
- Can intake against the gate
- Does not jam while vectoring

Vectored Intake Wheels

Problem:
Since the vector rollers are made from PLA, the hard surface does not grip balls properly and the intake is slowed.

VERSION 1

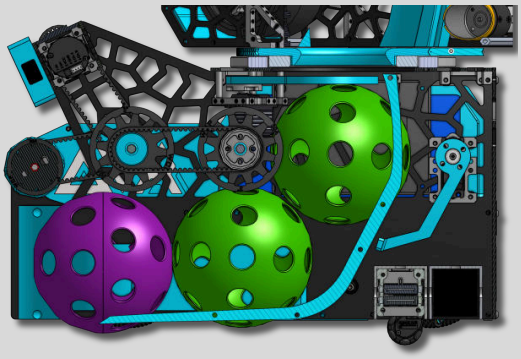


VERSION 2

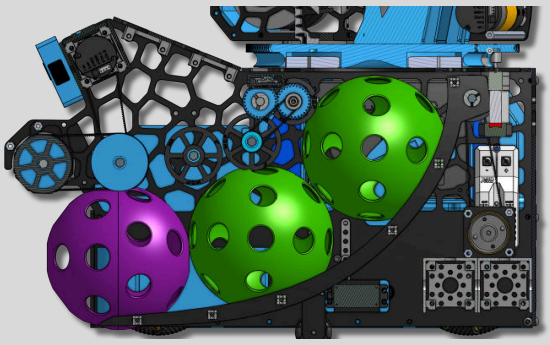


Solution:
We added small divots in each vector roller, and added a layer of surgical tubing to grip balls more and intake at a faster speed.

Transfer Iterations



VERSION 1



VERSION 2

Transfer Pros vs. Cons

PROS:
- Consistent transfer times
- Easy to repair/make changes

CONS:
- Frequently Jams
- Inconsistent compression
- Slow transfer time (0.3 sec)

PROS:
- Never jams
- Consistent compression
- Fast transfer time (0.2 sec)
- More contact with balls

CONS:
- Difficult to repair
- Complex to assemble

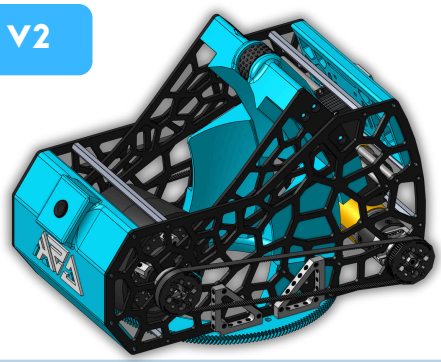
As one of the more **complex** subsystems on our robot, we have designed **28** CAD iterations and 3 major prototypes/versions of our shooter **demonstrating** our **relentless** effort to follow the **PRIME Engineering Process**.

V1



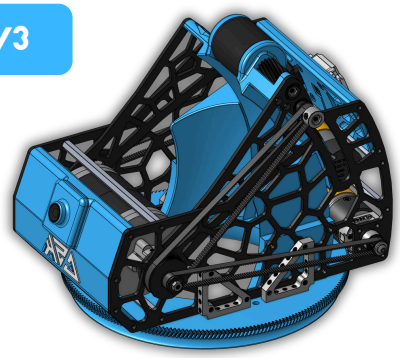
- Good artifact grip from **30A** durometer solid-core wheels
- Simple and **reliable** fixed hood design
- Compact motor packaging for **easy turret integration**
- **Lightweight pocketed sideplates** reduce turret inertia
- Fixed hood limits shooting range flexibility
- Low **flywheel inertia** causes RPM drop during bursts

V2



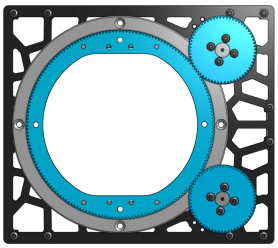
- **High flywheel MOI** (6604 g·cm²) improves shot consistency
- Adjustable hood enables shooting from **anywhere** on the field
- **Faster recovery** when firing multiple artifacts **consecutively**
- **Counterrollers** reduce artifact backspin
- Increased mass slows turret rotation
- Longer spin-up time due to high **rotational inertia**

V3



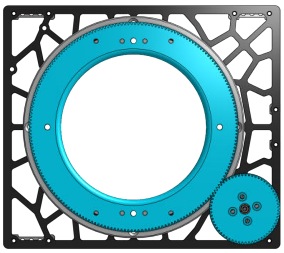
- **Faster flywheel** spin-up time
- Minimal RPM drop for consecutive shots
- Compact design improves **turret responsiveness**
- Powered counterroller (2.3:1) actively **prevents backspin**
- Eliminates bounce-outs and **improves scoring reliability**
- High flywheel MOI (7444.4 g·cm²) **improves shot consistency**

VERSION 1



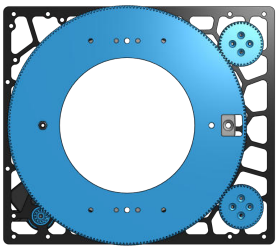
- **2× Axon MAX+ Servos**
- High torque output enables **fast turret acceleration**
- Dual-servo configuration provides **strong holding torque** under load
- **Compact servo packaging** simplifies initial turret integration
- Built-in servo control allows straightforward position commands
- **Servo synchronization** drift can reduce long-term accuracy
- Slow turret movement

VERSION 2



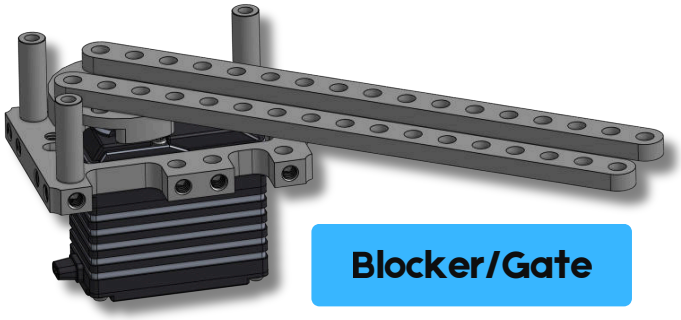
- **1× GoBilda 312 RPM Motor**
- **Encoder feedback** enables precise and repeatable turret positioning
- Single-motor design simplifies **mechanical alignment** and wiring
- Improved **positional accuracy** compared to servo-based designs
- Lower peak torque than **dual-servo configuration**
- Larger motor footprint increases turret **packaging constraints**

VERSION 3



- **2× Melonbotics 1000 RPM CR Servos**
- High rotational speed enables **rapid target acquisition**
- External REV through-bore encoder provides accurate **absolute position feedback**
- **Dual CR** servos distribute load and reduce mechanical stress
- Lightweight and compact design improves turret **responsiveness**
- Encoder-based feedback increases **repeatability** and **reliability**

Blocker/Gate



- **Servo-driven gate** prevents artifacts from entering the shooter until commanded
- Provides precise **timing control** for shooting sequences
- Simple rigid arm reliably blocks artifacts without **complex geometry**
- Prevents **premature feeding**, improving shot consistency
- Compact design adds **minimal mass** to the shooter
- Simple mechanism improves **reliability** and **durability**

PRIME is our structured engineering design process that guides how we approach building and improving our robot. It stands for Problem, Research, Ideate, Make, and Evaluate. This framework helps us think critically, stay organized, and make decisions based on strategy rather than impulse. Instead of jumping straight into building, we follow a clear sequence that encourages planning, analysis, and iteration. PRIME ensures that every subsystem on our robot is intentional and aligned with our game strategy. By continuously cycling through this process, we maintain consistency, adaptability, and steady improvement throughout the entire FTC season.

P – Problem

Define the right challenge before building

- Break down scoring priorities
- Analyze time + match constraints
- Identify mechanical limits
- Align robot goals with game strategy





R – Research

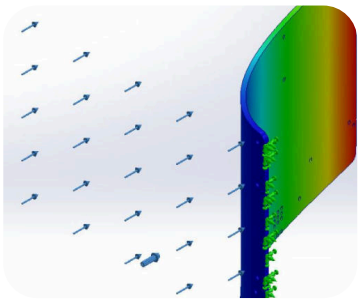
Gather data before making decisions.

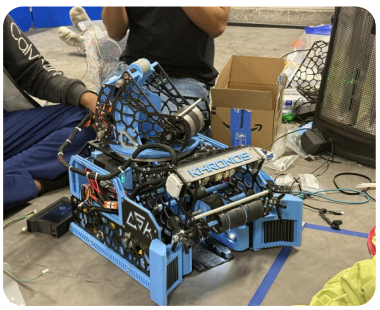
- Study past FTC solutions
- Prototype key mechanisms early
- Test subsystems independently
- Consult mentors for feedback
- Compare tradeoffs objectively

I – Ideate

Turn research into strong concepts.

- Brainstorm multiple subsystem designs
- Sketch and CAD alternatives
- Debate pros and cons
- Optimize for efficiency + reliability
- Select designs that maximize scoring





M – Make

Transform concepts into performance.

- Rapid prototype + iterate
- Manufacture and assemble parts
- Integrate hardware + software
- Refine wiring and code structure
- Improve consistency through testing

E – Evaluate

Test, measure, improve.

- Run controlled practice sessions
- Simulate competition pressure
- Track cycle time + accuracy
- Identify weak points
- Redesign for peak performance



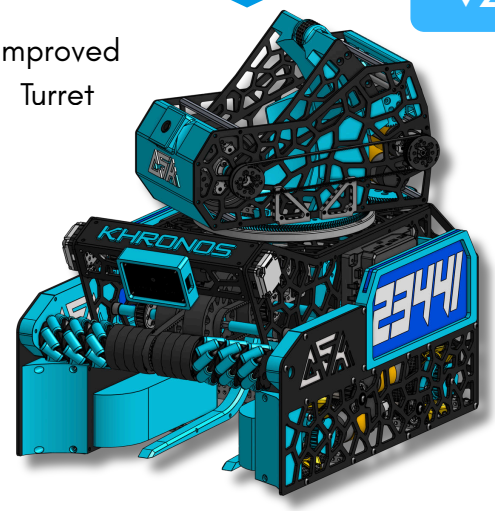
Robot Evolution

V1



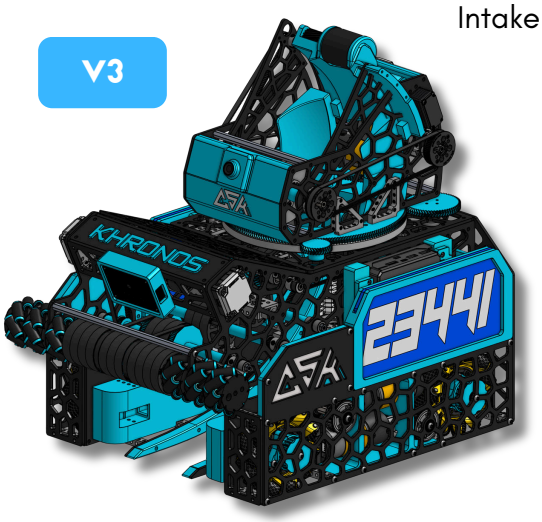
V2

Improved Turret



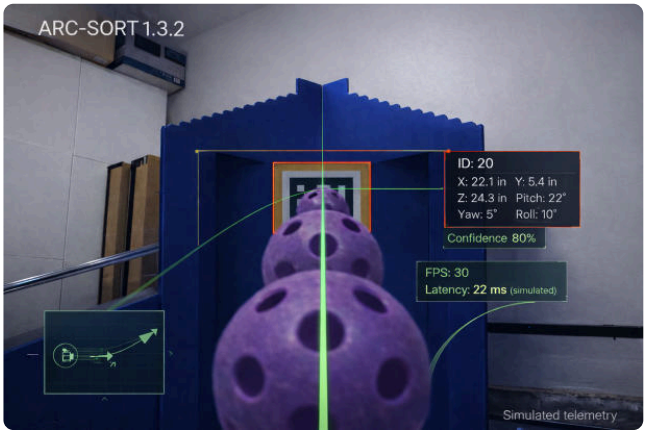
Improved Intake

V3



Team Created CameraVision Method

This year, recognizing **camera vision's** critical role in the DECODE season, our team leveraged online FTC resources to integrate it into our **codebase**. We evaluated multiple vision libraries, but encountered **unreliable pose estimates** and **inconsistent AprilTag detection** under variable lighting. After in-depth research into FTC VisionPortal capabilities, we developed **ARC-CV (Autonomous Range & Curve Computer Vision),** our custom vision layer built atop the official SDK. ARC-CV provides driver-assist aiming by processing AprilTag detections to estimate robot pose relative to scoring targets, while analyzing **simulated** ball trajectories to compute path errors. It factors in **detection confidence** alongside horizontal/vertical miss distances, feeding these metrics into real-time adjustments for our shooter's **angle and velocity** equations. All telemetry overlays are clearly marked as prototype/simulated during **development**.

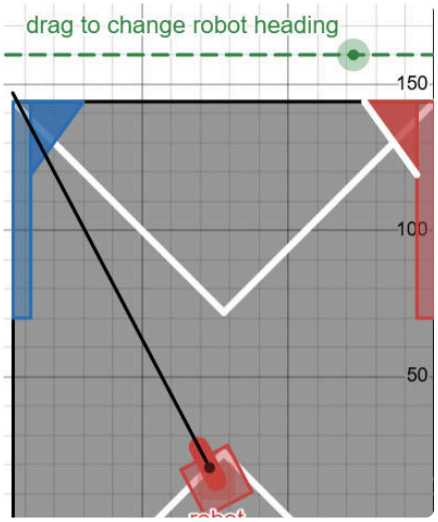


Iterative Testing and Validation

We validated ARC-CV through controlled lighting tests, autonomous repetition trials, and empirical shooter calibration. By logging detection confidence, latency, and miss distances, we refined polynomial fits and trajectory adjustments to reduce drift and increase scoring consistency under variable field conditions.

The picture on the right shows a Desmos Graphing Calculator graph we created to solve for shooting input parameters.

Our localization evolved from **two-wheel odometry (parallel + perpendicular dead wheels**, Control Hub IMU heading) to **three-wheel** (two parallel + one perpendicular pod) for **direct heading calculation** and less drift. **Encoder slot limits** led us to the goBILDA Pinpoint computer.



Pinpoint fuses **two-wheel odometry with pre-calibrated IMU** (0.002% heading accuracy), matching **three-wheel precision** with two-wheel simplicity. We used Desmos to plot **empirical RPM/hood angle** vs. distance data, fitting **polynomial equations** for real-time shooter control.

ARC-CV uses **AprilTag** pose for driver-assist aiming, analyzes simulated **ball trajectories** for path errors, and adjusts shooter **angle/velocity** based on confidence and miss distances. All overlays marked **prototype/simulated**.

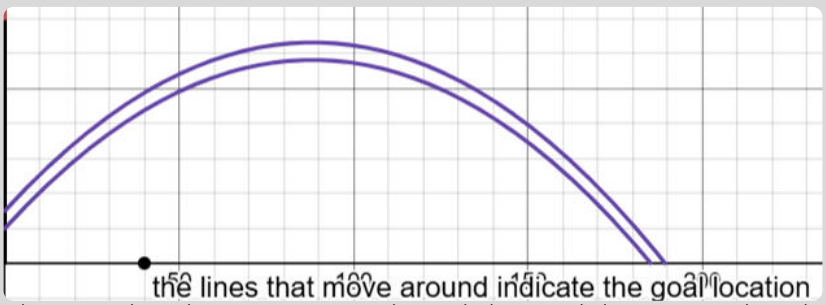
After deployment, we stress tested ARC-CV under varied lighting and field layouts to improve consistency. We optimized filtering, reduced latency, and fused tag pose data with odometry for precise autonomous alignment.

Translational Velocity Adjustment Deriving Equations

$$\Delta\bar{p} + (\bar{v}_t - \bar{v}_r)t = \bar{v}_m \bar{d}_t\, t + -\tfrac{1}{2}\bar{g}t^2, \quad \Delta\bar{p} = \bar{p}_t - \bar{p}_s$$

$$\vec{d}(t) = \frac{\Delta\bar{p} + +(\bar{v}_t - \bar{v}_r)t - \frac{1}{2}\bar{g}t^2}{\|\Delta\bar{p} + +(\bar{v}_t - \bar{v}_r)t - \frac{1}{2}\bar{g}t^{t2}\|}$$

$$v_m(t) = \frac{\|\Delta\bar{p} + +(\bar{v}_t - \bar{v}_r)t - \frac{1}{2}\bar{g}t^{t2}\|}{t}$$



The picture above depicts a Desmos Graphing Calculator graph that represents the path that the balls will have to take post-launch.

Simulated Artifact Pathing Equations

$$y = s_{height} + \left(v\sin\left(s_{angle}\right)\right)\left(\frac{x}{v\cos\left(s_{angle}\right)}\right) - \frac{1}{2}g\left(\frac{x}{v\cos\left(s_{angle}\right)}\right)^2 \{0 \le x\}\{y \ge 0\}$$

$$y = s_{height} + \left(v\sin\left(s_{angle}\right)\right)\left(\frac{x}{v\cos\left(s_{angle}\right)}\right) - \frac{1}{2}g\left(\frac{x}{v\cos\left(s_{angle}\right)}\right)^2 + 5\{0 \le x\}\{y \ge 0\}$$

$$d_{max} = \frac{v\cos\left(s_{angle}\right)}{g}\left(v\sin\left(s_{angle}\right) + \sqrt{\left(\left(v\sin\left(s_{angle}\right)\right)^2 + 2g\left(s_{height} - 38.500000\right)\right)}\right)$$

$$= 144.64680795$$

Our team has created a fully custom **vision pipeline**, **selection algorithm**, and **robot movement algorithm**, allowing us to cycle artifacts from any point on the field in autonomous.

Step 1 : Filtering for Artifact Color & Image Refinement

To start, we take a picture with our **LimeLight 3A** camera. With this image, we create a **color mask** in a region of interest. We also use **morphology functions** to reduce noise and focus only on specific desired artifacts.



Step 2 : Separating Individual Artifacts

Next, we apply **canny edge** detection and contour detection to generate **bounding boxes** around individual artifacts, separating them even when they appear **clustered** together.



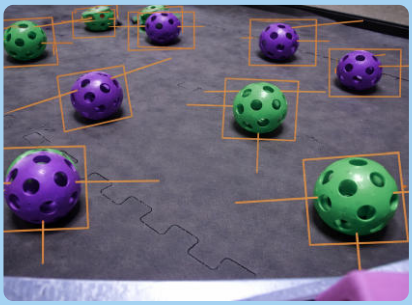
Step 3 : Apply Homography Transformation

Next, we perform a **homography transformation** to obtain **real world coordinates** and angle by **mapping** the real world points onto **pixel coordinates**.



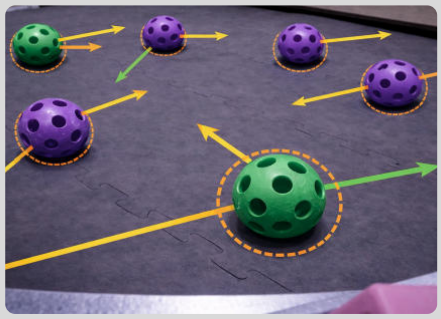
Step 4 : Eliminate Unreachable Artifacts

Next, for each artifact, we draw rectangles along the **perpendicular bisector** of the artifact and check whether they **intersect** any artifacts; if so, that artifact is removed from our **candidate list**.



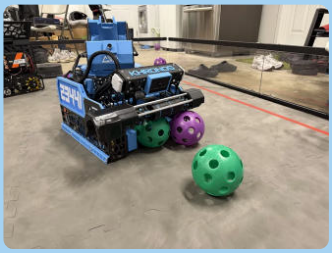
Step 5 : Selecting Artifact to Intake

For each artifact, we draw a **circle centered** at the intake that just **contacts** the ball, then construct a **tangent at that point**. We then select the ball whose tangent direction requires **the least physical rotation.**



Step 6 : Translate Position to Desired Artifact

Using the **real-world coordinates** of the chosen ball, we position the drivetrain and turret 2 inches **offset** from its **perpendicular bisector**. Then we drive forward, using our intake and robot position, until the **infrared sensor** detects the ball.



OLD Computer Vision: Blob Based Detection

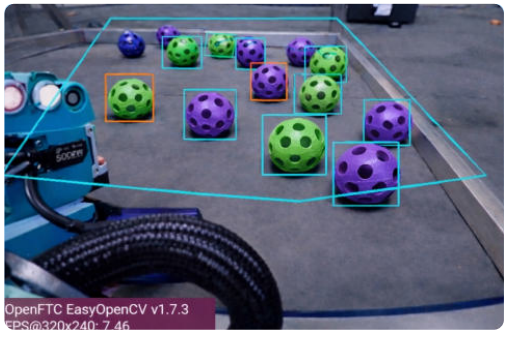
Throughout early DECODE events, our team used a **simpler vision pipeline** that detected a blob of balls using **contour detection** and just drove our intake into the blob. This approach was very **ineffective**, which pushed us to **redesign** for later events.

V1 - Blob Detection

- Simple to implement
- Cannot target individual balls
- Ignores ball approach angle
- Balls are treated as one large cluster
- Around 30% success rate

V2 - Blob Detection

- Individually targets balls to intake
- Computes real-world coordinates
- Uses intake alignment for optimal angle
- More complex to implement



Sensors & Automation

Sensors

Our robot uses **two color sensors** near the intake to detect and confirm **possession** of balls, and to differentiate between field elements when lining up shots. An **absolute analog encoder** tracks turret or hood angle with a true zero, so we never need to re-home between matches. **Motor encoders** on the drivetrain and key mechanisms provide precise **speed and position feedback**, improving path following and **shooter consistency**.



Localization

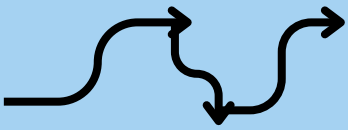
We localize using **odometry** pods feeding a dedicated odometry **computer**. This gives smooth, reliable pose **estimation** across all tiles for both **autonomous** and tele-op driving.



PATH FOLLOWING

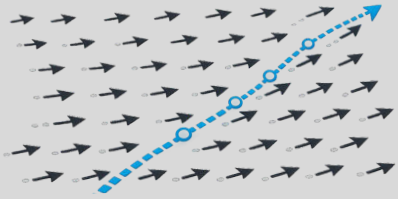
Movement V1

Initially, we drove using basic **driver control** and simple straight-line autos. This **limited** our consistency and made complex **multi-shot autos** difficult.



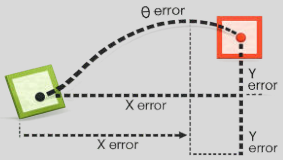
Movement V2

Next, we **adopted Road Runner** to generate predefined trajectories. This improved repeatability but could be **inefficient and slow to react** to sharp turns.



Movement V3

Finally, we **upgraded to Pedro Pathing** for fast, consistent motion. It **adapts for centripetal force**, allowing tighter, smoother paths at higher speeds.



Software Architecture

Instead of writing motor power values directly, we first store them. Each **device subscribes to a central publisher class**, which sends all commands to the hardware in one batch. This **ensures consistent device writes** and **reduces redundant per-loop updates**, improving efficiency and minimizing latency

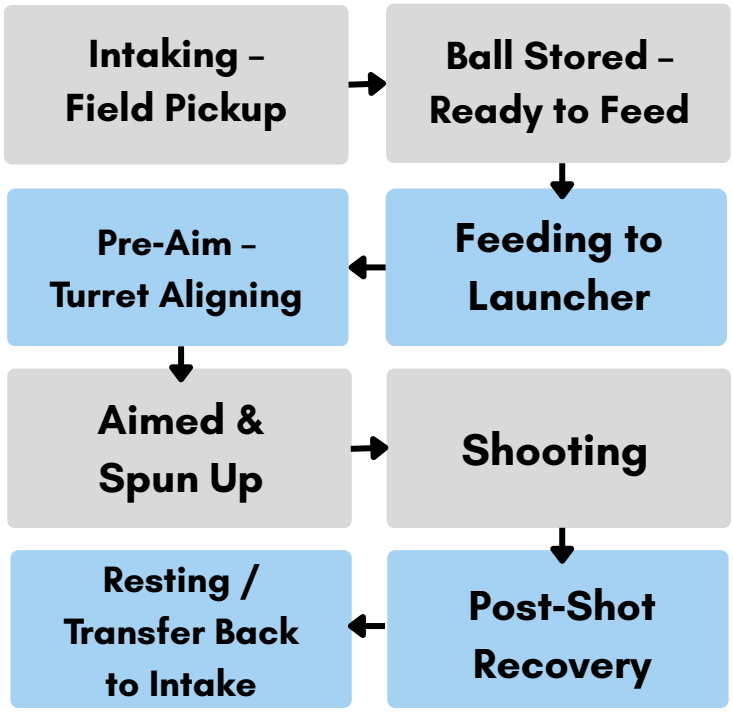


For autonomous, our team **built an actions system** to simplify creating and managing routines. We use a Java queue to **store pending actions and execute them** in order during auto. The following commands let us build complex paths and behaviors from simple building blocks:

- `executeUntilTrue()` runs an action until a condition becomes true
- `finishActions()` runs all queued actions without adding a new drive target

Robot State Machine

This season, our team uses a **custom state machine** to control the robot in both **auto and tele-op**. It handles all **subsystem** logic in the background, making autonomous routines easier to write and reducing **driver controls.**



The **CSK Foundation** operates as a registered **501(c)(3) non-profit organization**. This designation reflects our dedication to education and community service; every program ran is to **spread STEM literacy** and support our community, with all of our efforts focused on **impact over profit**.

Cost Breakdown

CATEGORY	DETAILS	COST
FTC REGISTRATION	FIRST Registration, SoCal Registration	$595
ROBOT EXPENSES	Motors, Control Hub, Expansion Hub, Sensors, Battery, etc.	$5000
	Raw Materials (Aluminum, Plastics, 3D Printing Filament)	$900
	Chassis + Wheels + Gearing	$1000
TOOLS & EQUIPTMENT	Field Tiles, Field Borders, New Season Field Elements	$600
	Allen Keys, T-Handles, Drills, Soldering Tools	$400
OUTREACH & MEDIA	Posters, Shirts, Event Fees, Website Domain	$600
MISCELLANEOUS	Safety Equipment, Team Jerseys, Pit Decorations	$800
WORKSPACE RENT	Working Area, Item Storage, Equipment Storage	$8000
TOTAL COST		$17895

We are incredibly thankful for **our sponsors**. Our team has earned these partnerships through proactive outreach and a **commitment to building relationships**. We love working with sponsors who recognize the passion we bring because it **motivates us** to perform at the highest level.

- Get **local businesses** involved as they are a **direct representation of our community** and with their support it motivates us even more
- Prepare the **CSK Challeng3rs & CSK Robot3rs** by providing some resources of our own and teaching them how to receive funding and **spend money wisely**

Thanks Sponsors!

We make sure to be very **proactive** and frequently **maintain a strong relationship** with them to ensure our **mutual**, **long-term success**.

Sponsor / Grant	Contribution ($)
CSK Foundation	$4,000
Gene Haas Foundation	$2,000
4aiSoft	$2,000
Softbox LLC	$1,000
Intuitive Foundation	$1,000
Onshape	$500
REV Robotics	$500
goBILDA	$500
CNC Madness	Utilities
Total Funding	**$11,500**

Risk Mitigation

Risk	Impact	Score	Mitigation
Shipping delays	Prevents our team from repairing our robot and progressing	High	Reach out to sister teams or access past supplies to improvise
Lower funding from prior sponsors	Finances may limit new opportunities	Medium	New fundraisers identifying new sponsors and always have spare $
Purchases needed; but no budget	Without a budget for immediate purchases, we get slowed down	High	Communicate with our parents to help us allocate some money
Out of Stock Items	Without even an order placed, there isn't anything to anticipate	High	Establish clear roles, and check back in with our coach

Incentive Sponsorship Packages

To help attracting partners, our team developed a comprehensive sponsorship package, using a structured list to determine specific perks for our supporters. From digital recognition on our website to prominent logo logo placement on our competition robot and team apparel, incentives ensure that our sponsors have a clear vision of the tangible benefits they receive. By providing transparency to our sponsors they can choose a level of support best aligns with their goals while directly fueling our technical innovation and community outreach.



Fundraising Events





We regularly take our robot on the road to **engage with the community** and showcase the power of hands-on engineering. By demonstrating our machine at large public venues and youth workshops, we **build the connections necessary to fund our season** and keep our team moving forward. Every interaction serves as an opportunity for us to bridge the gap between robotics and our community, **turning curiosity into future engineers**!

Team Sustainability



Coach Chandra provides the team with guidance by hosting regular meetings, getting all work done **behind the scenes**, and making sure we **meet deadlines**.





Captains help relay messages to other members






Flow of Information and Structure

- **Coach Chandra** provides our team with essential organization by making a structured schedule and holding all members accountable for each meeting and event
- Using their professional experience, our **industry mentors** aided us in problems we encounter while building and coding as well as logistical dilemmas
- Our **captains** relay information from others, and lead our team everyday through in person meetings and online through text chat and voice calls

At **CSK Foundation**, we hope to carry on our foundation's legacy and continue to inspire and empower children through STEM. Rather than chasing immediate success, we take greater pride in **shaping the next generation to achieve even more**. We do this by setting aside time not only for our teams progression but also to train and mentor younger students with our coaches and industry mentors. With early exposure, our goal for the younger generation is nothing but success.

Extending The CSK Legacy

Currently we have many students apart of First Lego League (FLL) that we are actively mentoring to prepare them for the step up into FTC. To do so, **we made a short educational guide** that we ask each student to read to learn the basic guidelines of FTC.

Currently we have many students apart of **First Lego League** (FLL) that we are actively mentoring to prepare them for the step up into FTC. While both leagues are apart of FIRST, there is a big step up from legos to metal and from blocks to java. To ease them into adusting, **we made a short educational guide** that we ask each student to read and analyze to learn the basic guidelines of FTC and how to lead a successful team

Leading by Example

FTC SoCal ILTs: **1st Place Design Award Winner**



CSK Robot3rs: Won 2nd Place Robot Design overall

CSK Challeng3rs: Won 1st place Overall Robot Performance



CSK Robot3rs



CSK Challeng3rs

The Aspiring Developers



Some things we emphasize to the students in the **Aspiring DEVELOP3RS** FTC Guide:

- **Gracious Professionalism**
- A **tentative pacing plan** to prevent them from being overwhelmed
- Financial Literacy and **being frugal** are important to maximize resources
- **Building/CADing** (Onshape guide) as well as how we source our parts from various sponsors and FIRST suppliers (ex: CNC Madness, GoBilda, REV and more)
- **Coding** (Java) Using Android Studio and Java to speed up the coding process from block coding

Member	Individual Goal	Steps taken and Contribution to the team
Suryaa	Advance CAD & leadership skills	Utilized CAD to design the robot, took charge for meetings
Pranav	Gain coding skills & mentor experience	Drastically improved robot during tele-op and led meetings
Sachin	Develop mechanical & CAD expertise	Dealt with complex subsystems & CADed intake system
Anish	Advance mechanic skills & problem solve	Optimized robot wiring and built core robot parts
Aahan	Develop skills in vision & auton code	Explored & integrated Limelight in code using OpenCV
Krish	Apply critical thinking to mechanics	Executed tough mechanical work without an easy solution
Ankit	Build technical skills & work ethic	Determined to come to every meeting and built main parts
Roshan	Work on analytical skills & experience	Worked on multiple subsystems, and learned basic mechanics